



AWS Runbook Generator

User Guide

Version 1.0 — June 2026
Workmates Core2Cloud Solution Ltd.

Internal — Team Distribution

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1. Introduction

1.1 Purpose

The AWS Runbook Generator is an automated infrastructure documentation tool that discovers AWS resources, captures live console screenshots, generates AI-powered descriptions, and produces a professional Microsoft Word (.docx) runbook document — all in a single command.

1.2 Key Features

- Automated AWS Resource Discovery — Scans resources by VPC or Tag across 20+ AWS service categories
- Live Console Screenshots — Captures actual AWS Console views of each resource using headless Chromium
- AI-Powered Descriptions — Generates intelligent descriptions for each infrastructure section using LLM
- EC2 Application Discovery — Discovers running services, ports, and configurations on Linux instances via SSM
- Professional DOCX Output — Produces a branded Word document with cover page, table of contents, architecture diagram, and resource sections

1.3 Supported AWS Services

Category	Resources Discovered
Networking	VPC, Subnets, Route Tables, IGW, NAT Gateway, EIPs, VPC Peering, VPC Endpoints
Compute	EC2 Instances, Lambda Functions
Containers	ECS Clusters/Services, EKS Clusters
Load Balancing	ALB/NLB, Target Groups
Database	RDS Instances/Clusters, DynamoDB Tables, ElastiCache
Storage	S3 Buckets
Security	Security Groups, IAM Users, KMS Keys, ACM Certificates, Secrets Manager
Monitoring	CloudWatch Log Groups, Alarms, SNS Topics, SQS Queues
Cost	AWS Budgets
CDN & DNS	CloudFront Distributions, Route 53 Hosted Zones

2. Prerequisites

2.1 System Requirements

Requirement	Minimum	Recommended
Operating System	Ubuntu 20.04 / WSL2 / macOS 12+	Ubuntu 22.04+ / macOS 14+
RAM	2 GB	4 GB
Disk Space	500 MB	1 GB
Network	Internet access required	Stable broadband

2.2 AWS Requirements

Requirement	Details
AWS Account	Active AWS account with resources deployed
IAM User	Programmatic access (Access Key + Secret Key)
Permissions	Read access to target services + sts:GetFederationToken
SSM Agent	Installed on EC2 instances (for application discovery)

2.3 Required IAM Policy

Attach the following IAM policy to the user that will run this tool:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "ec2:Describe*", "elasticloadbalancing:Describe*", "rds:Describe*",
      "s3:ListBuckets", "s3:GetBucketLocation", "lambda:ListFunctions",
      "dynamodb:ListTables", "elasticache:DescribeCacheClusters",
      "ecs:ListClusters", "ecs:DescribeClusters", "ecs:ListServices",
      "ecs:DescribeServices", "eks:ListClusters", "eks:DescribeCluster",
      "cloudfront:ListDistributions", "route53:ListHostedZones",
      "iam:ListUsers", "iam:ListRoles", "kms:ListKeys", "kms:DescribeKey",
      "sns:ListTopics", "sqs:ListQueues", "logs:DescribeLogGroups",
      "cloudwatch:DescribeAlarms", "budgets:DescribeBudgets",
      "acm:ListCertificates", "secretsmanager:ListSecrets",
      "ssm:SendCommand", "ssm:GetCommandInvocation",
      "ssm:DescribeInstanceInformation",
      "sts:GetCallerIdentity", "sts:GetFederationToken"
    ],
    "Resource": "*"
  }]
}
```

Important: The `sts:GetFederationToken` permission is critical — it enables the tool to authenticate into the AWS Console for capturing screenshots.

3. Installation

3.1 One-Command Installation (Recommended)

Linux / Mac / WSL:

```
curl -sL https://workmates-runbook-tool.s3.ap-south-1.amazonaws.com/setup.sh | bash
```

Windows (PowerShell):

```
irm https://workmates-runbook-tool.s3.ap-south-1.amazonaws.com/setup.ps1 | iex
```

This single command performs the following:

- Checks for Node.js ≥ 20 — installs Node.js 22 LTS if missing
- Downloads the tool package from S3
- Extracts to `~/runbook-tool/` (Linux) or `%USERPROFILE%\runbook-tool` (Windows)
- Installs Node.js dependencies
- Installs Playwright Chromium browser
- Automatically launches the interactive CLI

3.2 Manual Installation

```
# Step 1: Download and extract
curl -O https://workmates-runbook-tool.s3.ap-south-1.amazonaws.com/runbook-tool-latest.zip
unzip runbook-tool-latest.zip -d ~/runbook-tool
cd ~/runbook-tool

# Step 2: Install dependencies
npm install

# Step 3: Install Playwright Chromium
npx playwright install chromium

# Step 4: Launch
npm start
```

4. Usage

4.1 Launching the Tool

```
cd ~/runbook-tool
npm start
```

4.2 Interactive Configuration

The tool will prompt you for the following information:

Prompt	Description	Example
AWS Access Key ID	Your IAM user access key	AKIA...
AWS Secret Access Key	Your IAM user secret key (hidden)	*****
AWS Region	Target AWS region	ap-south-1
Discovery Mode	tag (by resource tag) or vpc (by VPC ID)	tag
Tag Key	Tag key to filter resources	Project
Tag Value	Tag value to filter resources	My-Project
Project Name	Name displayed on the runbook cover	My-Project
Company Name	Company name on cover page	Workmates Core2Cloud Solution Ltd.
Created By	Author name	John Doe
Environment	Environment label	Production

4.3 Pipeline Stages

Stage	Description	Duration
1. Discovery	Scans AWS resources via APIs	30–60 seconds
2. URL Mapping	Builds AWS Console URLs	5–10 seconds
3. Screenshots	Captures console screenshots	5–15 minutes
4. SSM Discovery	Discovers apps on EC2 instances	30–60 seconds
5. DOCX Generation	Assembles the final document	10–30 seconds

Total execution time: 7–18 minutes depending on resource count and network speed.

5. Discovery Modes

5.1 Tag-Based Discovery (Recommended)

Best for projects where all resources are tagged consistently.

- Scans all resources in the region matching a specific tag key/value pair
- Use when your infrastructure follows a tagging strategy (e.g., Project=MyApp)

5.2 VPC-Based Discovery

Best for project-based infrastructure contained within a single VPC.

- Scans all resources associated with a specific VPC
- Use when your project infrastructure lives in a dedicated VPC

6. Output

6.1 Generated Files

File	Description
runbook.docx	Final professional runbook document
screenshots/	Directory containing all AWS Console screenshots
resources.json	Discovered resources (intermediate data)
console-urls.json	AWS Console URLs (intermediate data)
discovery-data.json	EC2 SSM application discovery results

6.2 Runbook Document Structure

The generated DOCX includes:

- Cover Page — Company logo, project name, environment, date, author
- Table of Contents — Auto-generated section links
- Architecture Flow Diagram — Auto-generated Mermaid diagram
- AI-Generated Project Summary — Intelligent overview of the infrastructure
- Resource Sections (per category) with screenshots and details
- EC2 Application Discovery — Running services, ports, configurations
- Security Observations — AI-generated security notes
- Contact Information

7. Troubleshooting

Issue	Cause	Solution
sts:GetFederationToken denied	Missing IAM permission	Add sts:GetFederationToken to IAM policy
Screenshots show wrong account	Default AWS creds overriding	Unset AWS_PROFILE env variable
SSM fails for Windows	Linux-only discovery script	Expected — Windows instances are skipped
Playwright timeout	Slow network	Retry the command
Node.js >= 20 required	Outdated Node.js	Run setup.sh (auto-installs Node 22)

Re-running: The tool is idempotent — you can safely re-run it. Each run overwrites previous output files.

8. Security Considerations

- Credentials are not stored — AWS credentials are used in-memory only and never written to disk
- Read-only access — The tool never creates, modifies, or deletes any infrastructure
- SSM Commands — Only read-only system commands (netstat, systemctl, cat) run on EC2 instances
- Screenshots — Captured via a federated console session that expires after the run

For issues or feature requests, contact the Workmates Core2Cloud Solution Ltd. implementation team.